

WASP-49b

R-1197

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-49_191114 · created 2026-07-31T06:12:48+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458801.8501 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.141 +/- 0.04
Transit depth (Rp/Rs)^2	1.99 +/- 1.13 [%]
Orbital Inclination (inc)	84.13 +/- 2.88
Airmass coefficient 1 (a1)	0.36 +/- 0.19
Airmass coefficient 2 (a2)	0.54 +/- 0.24
Scatter in the residuals of the lightcurve fit is	54.85 %
Best Comparison Star	#1 - [245, 54]
Optimal Aperture	2.05
Optimal Annulus	5.5
Transit Duration (day)	0.073 +/- 0.044

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.141 ± 0.04 vs 0.116 (deviation 0.59σ)
Duration fit vs published	0.073 d vs 0.09 d (deviation 0.36σ)
Mid-time O–C	5.2 min
Depth SNR	3.3
Residual scatter	54.85 %

Light curve

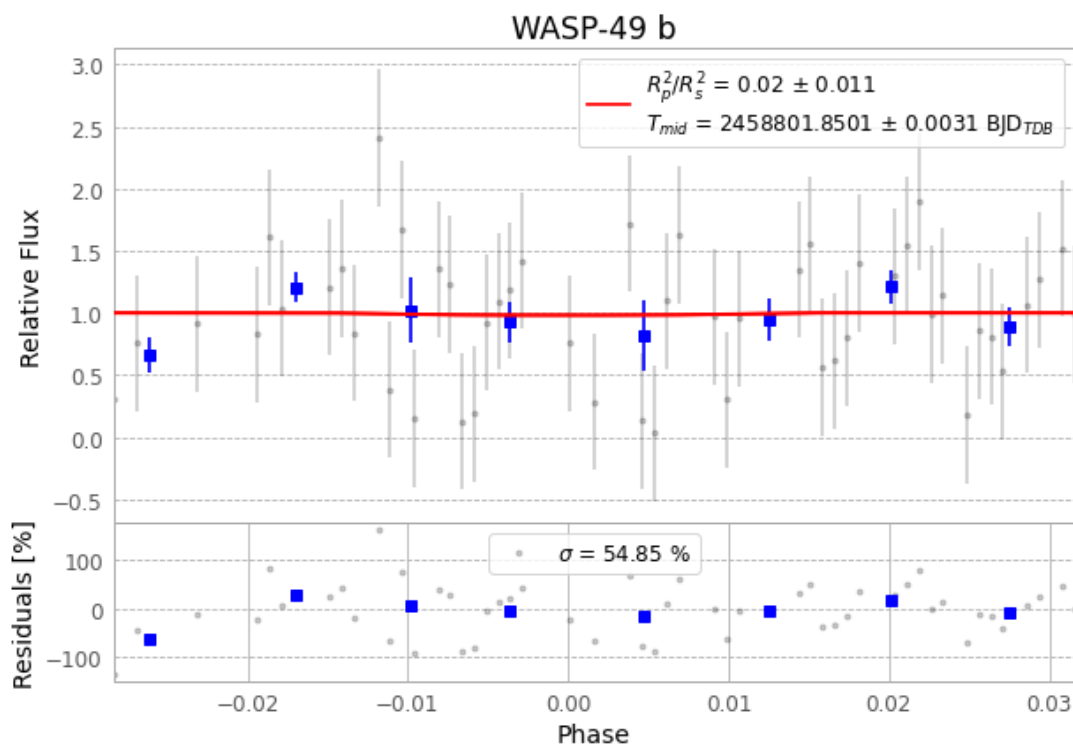


Figure 1 — FinalLightCurve_WASP-49 b_2019-11-13.png (SHA-256 10438efe87bf2d06...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [203, 160], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c7619332f5be05c5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

82 FITS frames (54 MB), WASP-49191114061812.FITS → WASP-49191114102412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-49-191114.log (deb79b075077...), FinalLightCurve_WASP-49 b_2019-11-13.png (10438efe87bf...), FinalLightCurve_WASP-49 b_2019-11-13.csv (d7fa0f7b287b...)

Compute resources

Wall time 135.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 06:12Z.